

Variable store

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.2 Programming fundamentals — Paper 2, Computational thinking, algorithms and programming
Lesson	Variable store (lesson 1 of the unit)
Lesson ID	pf-l01
Experience ID	pf-l01
Plan ID	PLAN-PF-L01
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Big question, outcomes and success criteria

Big question — How does a program store data, and how does it decide which line to run next?

Prerequisites

None within this unit: it is the first lesson. Treat the starter as a baseline check rather than retrieval of something you have taught.

Assessable outcomes

1. **pf-l01-o1** — know what variables and constants are, and how a value is assigned to each
2. **pf-l01-o2** — understand how input and output move data in and out of a program
3. **pf-l01-o3** — know what sequence, selection and iteration each do to the order a program runs in

Vocabulary, resources and preparation

Key vocabulary — Assignment, Count-controlled iteration

Resources

SLIDES	presentations/PF_L01_VariableStore_Lesson.pptx
WORKSHEET	worksheets/PF_L01_VariableStore_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=pf-l01

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.

3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on PF_L01_VariableStore_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Variables and assignment.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Variables and assignment

- A variable is a named location in memory holding a value that can change while the program runs.
- Assignment uses a single equals sign: the value on the right is stored in the variable on the left.
- In the OCR Exam Reference Language, $\text{score} = \text{score} + 10$ reads the old value, adds ten, then stores the result back.

Challenge on this wall: Explain why $\text{score} = \text{score} + 10$ is not a contradiction, even though both sides use the same name.

2. Constants

- A constant is a named value that is set when the program is written and cannot change while it runs.
- Constants make a program easier to read, and a value used many times can be changed in one place.
- A VAT rate or the number of lives a player starts with are good constants.

Challenge on this wall: Name two values in a quiz game that should be constants, and say why each one should not change.

3. Input and output

- Input brings data into a program, usually from the keyboard. Output sends data out, usually to the screen.
- In the OCR Exam Reference Language input() and print() do this work.
- input() always returns a string, so a number must be cast before it is used in arithmetic.

Challenge on this wall: Write three lines that ask a pupil for their name and age, then output a greeting using both.

4. Sequence

- Sequence means the statements are carried out one after another, in the order they are written.
- Change the order and you usually change the result.
- A variable must be given a value before any line that uses it.

Challenge on this wall: Explain why a program must input a price before the line that calculates the total.

5. Selection

- Selection chooses which statements to run by testing a condition that is either True or False.
- if ... then runs a block only when the condition is True; else runs when it is False; elseif tests a further condition.
- In the OCR Exam Reference Language the block ends with endif.

Challenge on this wall: Write selection that outputs Pass for a mark of 40 or more, and Fail for anything lower.

6. Iteration

- Iteration repeats a set of statements. A count-controlled loop repeats a known number of times.
- A condition-controlled loop repeats while a condition stays True, so the number of repeats is not known in advance.
- In the OCR Exam Reference Language a count-controlled loop uses for and next, and a condition-controlled loop uses while and endwhile.

Challenge on this wall: Give one job that suits a count-controlled loop and one that suits a condition-controlled loop, and say why.

Retrieval and the check for understanding

Starter — A program asks a pupil for their name and three test marks, then outputs their average. Write down every variable the program needs, and one value that would be better as a constant.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pf-101-s4-c1), so the wording a pupil sees matches the wording you say.

What does sequence mean?

Correct: Statements are carried out one after another in the order they are written

What a pupil might choose instead

- Statements are repeated until a condition is met
- A choice is made between two sets of statements
- Statements run in whichever order the computer finds quickest

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: Nothing is skipped and nothing is repeated: each line runs once, in order.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	13	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	1	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Variables and assignment	2	A variable is...
2	Constants	2	A constant is...
3	Input and output	2	input() always returns...
4	Sequence	2	Sequence means...
5	Selection	2	Selection is used when...
6	Iteration	2	A count-controlled loop... A condition-controlled loop...
final	Final challenge: three constructs	2	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
pf-101-x1	State the difference between a variable and a constant.	2	worksheet, Exam practice a); deck, Exam practice slide
pf-101-x2	Describe the difference between count-controlled and condition-controlled iteration. Give an example of when each would be used.	4	worksheet, Exam practice b); deck, Exam practice slide
pf-101-x3	A program asks for a password and keeps asking until the correct one is entered. Write an algorithm to do this, using the OCR Exam Reference Language or a language you know.	5	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pf-101.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What does a variable store?** — A value that is fixed and can never change; The name of the program
- **What does `total = total + 5` do?** — Checks whether `total` is equal to `total + 5`; Creates a second variable also called `total`
- **What is a constant?** — A variable with a very short name; A value stored on the hard disk
- **Which of these is best stored as a constant in a game?** — The player's current score; The player's name
- **What data type does `input()` give the program?** — An integer; A Boolean

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- The difference between a variable and a constant
- How input and output work, and why input must be cast
- Sequence, selection and count- and condition-controlled iteration

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is pf-102, Type foundry. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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